



Home Robotics: The Innovation Continues

Robots have become advanced enough to penetrate homes. The market for consumer robots has grown at a CAGR of seventeen per cent between 2014 and 2019



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Robots are everywhere today, from industries and medical systems to defence and agriculture sectors. There are millions of robots in the world today and more are being produced every day. But did you know that robots are slowly entering and becoming popular in homes as well?

Gamma chef robot is specialised in one-pot meals. And, Moley robot can create more than 2000 dishes designed by master chefs.

Tertill, developed by Franklin Robotics, is a solar-powered weeding robot used in gardens. It identifies weeds and pulls them out without using dangerous chemicals.

Toilbit cleans toilets. This robot uses cleaning capsules and a flexible brush that can reach everywhere in the bowl.

Buddy is an open source robot for homeowners to create their own programs and tasks for it.

These are just a few examples of home robots. A home robot, also called a domestic robot, is a type of service robot that is primarily used for household functions, but it may also

be used for education, entertainment or therapy purposes.

There are several household functions that are repetitive and labour-intensive. Some such home functions are cleaning, cooking, house maintenance, shopping, laundry and paying bills. Home robots are the key enabler for bringing about changes that can make a positive impact on such domestic functions.

While most domestic robots are simplistic, some are connected to home Wi-Fi networks or smart environments, and are autonomous to a large degree.

Types of domestic robots

Domestic robots are of the following types:

Indoor robots. These work around and inside homes. These can clean floors using sweeping and wet-cleaning functions. Some use wet-cleaning products like Swiffer, while others use disposable cleaning clothes for dry sweeping or reusable micro-fibre clothes for wet cleaning.

Ironing robots can be used to dry and iron clothes using hot air. Advanced ironing robots, such as Laundroid, use image analysis and artificial intelligence (AI) for this purpose.

Effie is an ironing robot that can iron twelve clothes at once. And, FoldiMate can fold and organise clothes.

Atlas is a robot built to perform such household tasks as sweeping, opening doors, climbing stairs, etc.

Cat litter robots self-clean litter boxes automatically and filter compact mass out into a built-in waste receptacle lined with an ordinary plastic bag. These are fully-automatic, and shift waste within minutes to



Effie can iron clothes (above); FoldiMate can fold and organise clothes (right)

